

Vedrana Ivezić

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Ph.D student (4th year) in Medical Informatics at UCLA with experience developing representation learning frameworks for multimodal biomedical data, including digital cytology, wearables, ultrasound, and digital pathology. Demonstrated expertise in Python, PyTorch, and large-scale self-supervised learning. Proven track record in building scalable AI systems for healthcare applications, with a focus on developing and improving foundation model generalization and robustness in low-resource clinical settings. Strong publication record in top-tier venues such as CVPR, MICCAI, and JAMIA.

EDUCATION

University of California, Los Angeles, CA

PhD student in Medical Informatics

Sept. 2022 - May 2027

(*expected*)

Princeton University, Princeton, NJ

B.A. Computer Science

Sept. 2018 - May 2022

GPA 3.74

Minor in Quantitative and Computational Biology

Magna cum laude

PUBLICATIONS

**Co-first authors. Please see more on [Google Scholar](#).*

EXPERIENCE

PhD Student, Medical Informatics - UCLA

Sept 2022 - Present

Biomedical AI Research Lab, University of California, Los Angeles

- Developing a novel automated multi-modal thyroid cancer diagnostic framework to aid clinical workflows
- Engineering self-supervised learning frameworks for interpretable and precise cytology image diagnostics
- Pioneering a multi-modal physiological foundation model by aligning temporal physiological patterns with the electronic health record for personalized health interventions
- Leading collaborations (e.g. Mayo Clinic) to ensure clinical utility and biological relevance of models

Human Factors (UX) Research Intern

Jan. - June 2026

VULVAI

- Synthesizing qualitative insights from beta-testing to inform the iterative development of the AI-driven health interface
- Optimizing data acquisition protocols to collect high-fidelity datasets structured for future multi-modal diagnostic AI models

Undergraduate Researcher

Sept. 2020 - May 2022

Troyanskaya Lab, Princeton University

- Engineered cell type-specific Bayesian networks to enhance functional gene prediction accuracy
- Leveraged machine learning frameworks identify molecular drivers of kidney disease
- Benchmarked cell type prediction methods across single-cell and single-nucleus RNA-seq datasets

Research Intern

June 2021 - April 2022

Jarosz Lab, Stanford University

- Analyzed FASTQ sequencing data from yeast strains to identify gene essentiality and construct gene interaction networks
- Applied statistical methods to evaluate gene correlations associated with prion phenotypes

PROJECTS

Cytology foundation model

Feb. 2025 - present

- Developed the first cytology foundation model using self-supervised learning
- Engineered a 1.4M image dataset using a custom pipeline to automate data cleaning and identify high-utility biological patches from public and private sources
- Outperforms natural image and histopathology foundation models on classification of cervical cancer (+3% accuracy) and cervical cell types (+4% accuracy)
- Captures unique morphological features with robust generalization to unseen multi-center datasets

Fitbit foundation model

Feb. 2023 - present

- Led multi-modal data engineering and statistical analysis for a heart failure randomized controlled trial
- Evaluated signal patterns from Fitbit and symptom-tracking survey data to characterize feasibility of longitudinal wearable data as a reliable clinical intervention for remote patient monitoring
- Developing physiological phenotyping models for amenorrhea from physiological data and clinical measurements in collaboration with the Mayo clinic
- Developing the first Fitbit foundation model for optimized representation learning of physiological patterns in sample limited datasets

SKILLS

Deep learning	Self-supervised training, contrastive learning, multi-modal learning, transformers, CNNs
Technical	Python, Pytorch, pandas, SQL, numpy, Scipy, scikit-learn, Git, openCV, OpenSlide

Last updated: January 5, 2026